Decision Memo

Q. Why a JEPA-style predictive latent for this problem, and what does it buy over the simpler peers you considered (x(t) as the latent, a plain Neural-ODE, a supervised decoder)?

The idea behind the predictive latent is that it contains information on per-patient progression rates that cannot be learned by the eight observations. Two patients with the same current status and different vulnerability move away from each other, and the direct predictor observes the same inputs and needs to average them. We compare the JEPA model to three competing models. The first is a direct recurrent next-state predictor where x(t) denotes the latent variable; it is our baseline model, and we equip it with the same constraint head as the JEPA in order to isolate the influence of the latent variable. The Neural-ODE provides the notion of continuous time which is not necessary on a fixed monthly grid and integrates to residual recurrent networks after discretization at one-month steps.

What it actually buys is the interesting part, because it is not what we expected. The baseline model is roughly thirty percent better (with MSE 0.0074 against 0.0106) even for the rapidly progressing group where we thought that using the latent was going to improve things, which does not happen. Indeed, with perfect observability, the encoder latent of the baseline decodes the same rate, and it is inefficient for the predictor to have to make its predictions via a bottleneck that only includes sixteen dimensions. The latent has earned its right of being observable when there is noise. Since the encoder latent encodes the entire sequence of observations up until now, the head of a very few dimensions decodes a de-noised estimate of the current state, and the ratchet rollout can rely on the decoded state rather than a single observation. The de-noised state is robust to noise; its error (0.082) is constant for all noise levels and beats the baseline near sigma 0.15 (0.082 against 0.106 at sigma 0.20). It is clear that the predictive latent purchases robustness, not accuracy in a clean data regime where the single generator does not have anything to do.

Q. What mechanism prevents representation collapse in your latent, and what metric would catch it if it crept in?

The latent can collapse into a useless representation where all patients are mapped to a single point. In this case, prediction becomes trivially perfect because the model is merely predicting a constant, without learning meaningful patterns.

To prevent this, we use a stop-gradient on the target, which avoids the predictor and target drifting toward the same constant. However, this alone does not prevent encoder collapse. Therefore, we introduce two additional regularizers: one encourages sufficient variance across the 16 latent dimensions, and the other encourages these dimensions to capture distinct information.

We monitor collapse using the number of active latent dimensions, which ranges from 1 to 16. An alarm is triggered if this value falls below 3.

This monitoring proved necessary. Early in training, the value dropped from about 3.3 to 1.3 because the decoder was given the current state to build on, and that alone predicts much of the near future, causing it to ignore the latent representation. Strengthening the regularization terms resolved this issue, and the value eventually stabilized around 3.5 (8.86 across the full test set).

Importantly, a collapsed model often achieves the lowest training loss because the task becomes trivial. Therefore, model selection cannot rely solely on loss. Instead, we select checkpoints with the best state reconstruction among those maintaining a healthy latent representation (at least 3 active dimensions).

Q. How do you keep predicted trajectories on the manifold of valid states, and what does that cost?

We parameterize the decoder to produce deltas that pass through sign-fixing functions rather than penalizing constraint violations or performing a projection after generation. In the case of F, D, and P, the next state is set to be the current state plus softplus of the raw output, which never decreases. In the case of M, the increment is softplus times F times C times dt, which is non-negative and is zero whenever F or C is zero, so the F-times-C coupling is encoded functionally. S takes the monotone branch away from ERCP and the free branch at ERCP, and the fast fields employ a bounded sigmoid. Projection would provide the same guarantee but would have zero gradient on the active set, while the softplus ensures always positive gradient. Since the guarantee has to hold regardless of parameters, we validate it by testing on 1,560 states with random input values and standard deviation of the network's output set to five, with zero violations observed in both cases; the model reports the violation rate to be zero on held-out data. Zero violation rate does not guarantee on-manifold generation, so a discriminator trained on differentiating between real data and generated data that violates dynamics (AUC 0.97) evaluates both models at the real-data level. Cost of such an approach is limited expressiveness and decreased optimization speed of the slow fields along with the imposed, not learned coupling. If the true hazard was not F times C, the model would learn compensating rate dynamics.

Q. Where is the residual risk?

The biggest risk is a structural one. Generalization within a generator involves learning the generator’s update function, so our probes can’t distinguish between a world model and generator inverter. The model under-predicts rapid progressors by 1.33x the error of the typical patient, and the baseline exhibits the same disparity, so it appears to be an artifact of training data rather than the architecture itself. The counterfactual of administering UDCA six months ahead of time, verified by re-running the generator and verifying with a shared noise stream, recovers the right direction for all responders but under-predicts the magnitude of the effect, by about three orders of magnitude (-0.0001 versus -0.0445), indicating the model is an observational one without any interventional semantics. Long horizon rollout error increases by about 1.75x whereas the violation rate remains at zero; error on unseen patients is 1.12x error on seen patients.

Q. What would you do next?

The next step after having discovered the existence of this ceiling in one type of generator is creating another generator with different constants. We will use one of them for training, and another one – for testing. Next step will be adding interventional counterfactuals, as currently we are measuring only the fact that the correlational model happened to be faithful and the magnitude is where the value is at stake. We will implement hierarchical latent, which separates slow fields from fast fields, as the current implementation of the latent barely changes during the rollout and the recurrence does not care about it at all. Finally, we will add prediction uncertainty.

Q. Why did the model predict decompensation at month 30?

Since we use 12 months' worth of historical data for prediction of the next 12 months, "month 30" is a reference point at the end of the prediction period. It should be noted that the evaluation selects a patient that is most decompensated in the test dataset.

Consider a patient diagnosed with primary sclerosing cholangitis (PSC) with response to ursodeoxycholic acid (UDCA) and a susceptibility of 2.409, a clearly fast progressing one. The fibrosis (F) at the end of the observed history is 0.649. In 12 months it becomes 1.0, which is stage four of cirrhosis. The predicted value for F is 0.711, corresponding to stage three.

Thus, our model has captured the correct direction but has failed to predict the proper extent, which requires an explanation.

The continued development of disease is correctly identified, since it focuses attention on cholestasis (C) 0.264 and inflammatory activity (A) 0.233. These two values correspond to the logic used by the simulator in the sense that the rate of development of fibrosis is determined by A and C. Thus, the model looks at the proper signals.

The underestimated extent of fibrosis development happens because the prediction is almost not based on the history. For example, removing any field of the history makes prediction move by no more than 0.001, and the predicted latent moves by 0.001 during the entire rollout. Therefore, we conclude that the forecast is made based on the current state of the patient, not on the history. The model learns "fibrosis progression continues from its current state" significantly better than "the patient is progressing fast," so a fast progressor will receive the typical slope.

This behavior can be seen in all patients in the cohort with our susceptibility probe, as well as in the shortcut described above. Nevertheless, we report this problem instead of hiding it since a confidently wrong prediction in the case of a fast progressor is something that should be known to a clinician.